The reporting-rate axis should not be in the uncertainty grid

Two investigations into the tagged-fish reporting-rate penalty, read alongside the SC22 tag analysis. The setting that declines to correct for reporting inserts pseudo-data into the likelihood on **96 of 98** tag release groups. The setting that does correct trips a numerical guard on **2**. Both are flawed; they are not equally flawed, and they do not bracket anything.

96 of 112 named checks held · 100 drawn · 88 ran · **80 converged (MGC ≤ 1e-4)** · 168 fixed-parameter MFCL evaluations

Executive summary — in plain language

Tuna are tagged, released, and some are later recaptured and reported. The assessment must assume what fraction of recaptures actually get reported — the **reporting rate**. Right after release there is a settling-in period, the **mixing period**, during which tagged fish have not yet spread through the population, so the model is not supposed to fit their recaptures.

But those fish have still been caught, and they still have to be taken out of the tagged group. MFCL offers two ways to do that, and the 2026 ensemble uses both — splitting the models roughly evenly and treating the choice as a source of uncertainty. **That treatment is not defensible.**

Neither option is correct, and one is clearly worse

The option that *does not* correct for reporting — **exclude** — removes recaptures at face value, which silently assumes every recaptured tag was reported. The model's own fitted rates say that is untrue for most of the tagging data. Worse, it inserts terms into the fitting criterion whose best possible answer is always “reporting is 100%”. Those terms are not information about the fishery; they are an artefact of the setting, and they push the estimated reporting rates upward, away from what the independent tag-seeding studies say.

The option that *does* correct — **include** — divides each reported recapture by the estimated reporting rate before removing it. That can give an unbiased answer, but only if the reporting rate is about right, the release groups are big enough to be informative, and the scaled-up removal does not try to take out more tags than exist. When the last condition fails, MFCL quietly caps the removal, and nothing in the standard model output says so.

Why this matters for advice

Reporting rates and fishing mortality trade off. If the model believes more recaptures were reported, it needs less fishing to explain the tags that came back. So the choice moves the answer. All figures below are on the **34 + 46 = 80 models that met the assessment's own convergence criterion** (maximum gradient ≤ 1e-4), which turns out to be exactly the set carrying a retained parameter file:

CHANCE FISHING EXCEEDS THE LIMIT	CHANCE STOCK IS BELOW THE LIMIT	FISHING MORTALITY
20% vs **29%** **exclude** vs **include** models (reported ensemble: 24%)	**15%** vs **24%** **exclude** vs **include** (reported ensemble: 19%)	**-15%** F/F_{MSY} under **exclude**, adjusted for every other axis

*These risk figures reflect only **structural** uncertainty — the spread across grid cells (h, K, τ, M0, effort creep, and the RR flag) — each cell taken at its single best-fitting parameter vector. They do not fold in **estimation** uncertainty, the spread of plausible parameter values within a single fit. Folding that in would widen every probability reported here and could shift the two arms' numbers unevenly; see the note below the adjusted arm effects for what stands in the way of computing it now.*

The corrected setting is also the fragile one

Of the 50 models drawn for each setting, **16 of the 50 include models were lost** — either failing to build or failing to converge — against **4 of 50** for **exclude**. That is 32% against 8%, a factor of four. Convergence failure alone runs 17% against 2%.

This is a real cost of the corrected setting and it should be stated plainly: dividing observed recaptures by an estimated rate is numerically demanding, and MFCL does not always get there. It is also the reason the two arms are not evenly represented — the surviving **include** models are the ones where the correction happened to be well-behaved.

Including both does not bracket the answer

The usual reason to carry two options in an uncertainty grid is that the truth probably sits between them. That reasoning does not apply here. Assuming complete reporting *cannot mathematically* remove more tags than correcting for reporting does. The two options are one-sided, not bracketing: running both widens the spread of results without bounding fishing mortality or stock size from the direction that matters.

What we recommend

If models fitted to tagging data must inform advice, use include only. It is capable of being right under stated conditions; **exclude** is biased low unless reporting is genuinely complete, which it is not.

The MFCL manual recommends the opposite, on the grounds that the correction can be numerically unstable. **That recommendation trades a rare, cheap problem for a pervasive, expensive one.** We measured the trade directly: at the reporting rates the models actually estimate, the numerical guard costs at most **1.04 objective units** and engages on **2 of 98** release groups. The alternative contaminates **96 of 98** — about 48 times as many — on every single model.

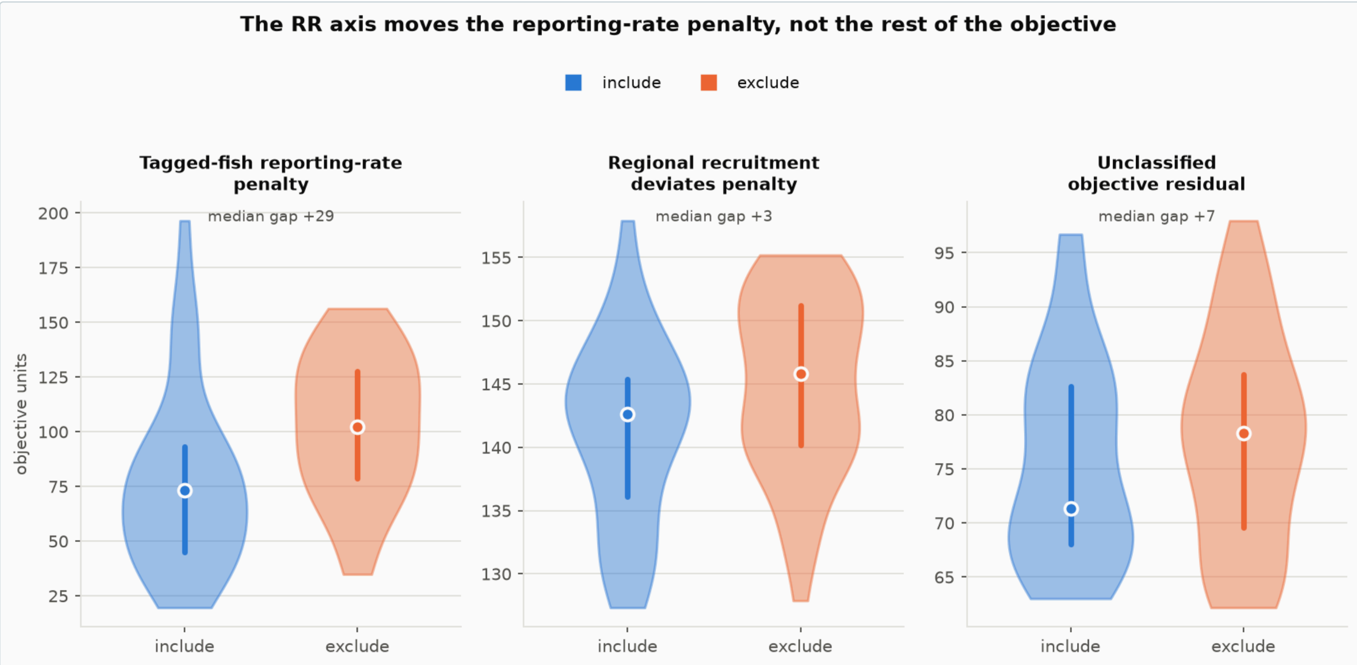
The deeper point, from the SC22 review, is that *both* options fix the tagged population deterministically from the observed returns, with no uncertainty and no test against what the model predicts. A large share of the tagging data steers the assessment without ever being fitted. The durable fix is to change MFCL so mixing-period recaptures are predicted from the tagged cohort and scored, with reporting rate and mixing-period availability estimated rather than assumed — or to base advice on models that do not fit tagging data at all.

The rest of this document sets out the evidence. Every directional claim names a check that either held or did not, and failures are reported in the same form as passes.

PART 1 · THE ORIGINAL QUESTION

Why the reporting-rate penalty moves with the axis

The starting observation: of all the objective-function components, the **tagged-fish reporting-rate penalty** is the one the `RR` axis really moves — median near 60 under **include** against 100 under **exclude**. The regional recruitment penalty differs by a couple of units; the unclassified residual barely differs. The effect is localised in the reporting-rate prior, not smeared across the fit.



Restricted to the 80 converged members. The `RR` axis moves the tagged-fish reporting-rate penalty by roughly ten times as much as it moves either of the other two components — the distributional picture behind the medians quoted above.

That penalty is a prior. Tag-seeding studies (SC22-SA-IP-05) give expected reporting rates for the purse-seine fisheries in regions 2, 3-west and 3-east; the penalty charges the model for departing from them. Transcribed from `multifan-cl/src/callpen.cpp`:

```
penalty = Σ over unique reporting-rate groups:  w_g · ( X̂_g − target_g/100 )²
```

Reconstructing it from the PAR files reproduced the reported component to **4.9 × 10⁻¹⁰** across all 80 members. The arithmetic is settled and everything downstream rests on it.

HELD `task3.reconstruction-matches-reported`

max|resid| = 4.86e-10 (tol 1e-06); median resid = 2.32e-12

Three structural corrections to the original framing. **Five** groups carry an informative prior, not three — three distinct priors, each shared by an RTTP and a PTTP/pooled group. Group 18 is `PS.EAST.3`, not region 4. And the selector in the source is the penalty matrix, not the active-flag matrix.

Not natural mortality

M0 has a tighter relationship with this penalty than the flag axis does, and the 88 models are a selected subset, so M0 leaking in was the obvious worry. It is not: M0 is balanced across arms, and adjusting for it *increases* the axis effect from 33.98 to 34.3 while R^2 rises from 0.45 to 0.85.

> **HELD** `task4.M0-balanced-across-arms`
>
> standardised mean difference (exclude - include) = -0.076

> **HELD** `task4.adjusted-arm-effect-positive`
>
> adjusted arm effect (exclude - include) = 35.54 [28.72, 42.36]

One balance check did not hold — K is imbalanced at K = 0.20 (include 7/41, exclude 16/47). K is in every adjusted model and the adjusted estimate is the larger one, so the imbalance does not manufacture the effect. Reported because it is real.

> **DID NOT HOLD** `task4.K-balanced-across-arms`
>
> largest level-share difference = 0.170

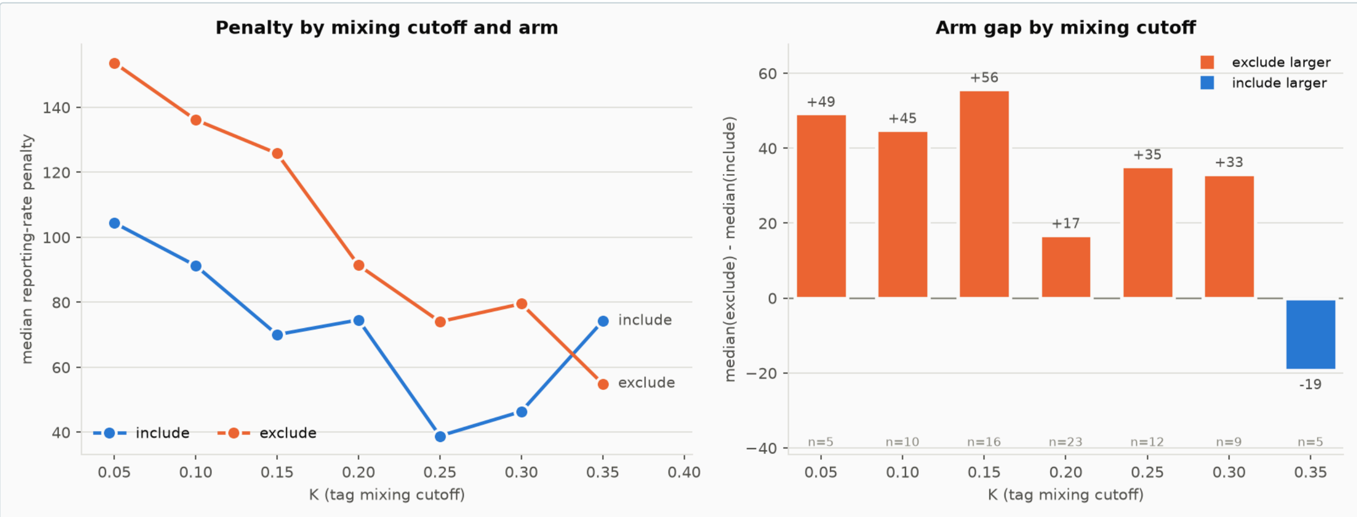
Which groups, and the mixing window

Group 17 dominates the penalty's *level* (68% of the include-arm median) but supplies only 19% of the arm *gap*; that comes from group 18 (45%) and group 7 (34%). Level and gap are answered by different parameters.

> **DID NOT HOLD** `task5.arm-gap-dominated-by-group-17`
>
> group 17 supplies 19.3% of the median total-penalty arm gap; group 18 supplies 45.0% and group 7 34.1%

The gap scales with the mixing window in both directions tested: arm × K = −218 [−298, −138], and arm × mean mixing period = +21.1 [13.7, 28.6].

> **HELD** `task6.arm-gap-shrinks-with-K`
>
> arm x K interaction on the raw penalty = -218 [-298.3, -137.7], p=7.74e-07



The gap is largest where mixing windows are longest and vanishes where they collapse — the first sign the mechanism lives inside the mixing period.

PART 2 · **THE MECHANISM**

Channel 2 — how exclude manufactures evidence for perfect reporting

The first investigation explained the gap by a *removal* argument: the reporting rate has an extra lever under **include**, so a smaller excursion suffices. That is correct but direction-agnostic. A second mechanism predicts the sign, and the observed sign is upward.

Mixing-period recaptures *are* scored under both settings — the likelihood loop starts at the release period and the multiplier never consults the flag. What differs is what the prediction is built from. MFCL solves for a fishing mortality that removes a target number of tags, and the flag sets that target. Writing `R` for observed returns and $\hat{\lambda}$ for the estimated reporting rate:

SETTING	REMOVAL TARGET	PREDICTED REPORTED RETURNS	CONSEQUENCE
include	`R / X̂`	`X̂ · (R/X̂) = R`	X̂ cancels — the term carries no information about it
exclude	`R`	`X̂ · R`	**minimised at X̂ = 1** — the term asserts perfect reporting

Under **exclude** every mixing-period recapture becomes a small vote for "reporting is complete", and the tag-seeding prior is the only thing pushing back.

Confirmed at frozen parameters

Take a converged model, freeze every parameter, and sweep one group's reporting rate under each setting. Nothing re-estimates, so nothing can compensate.

HELD `task1.exclude-mixing-block-monotone-toward-one`

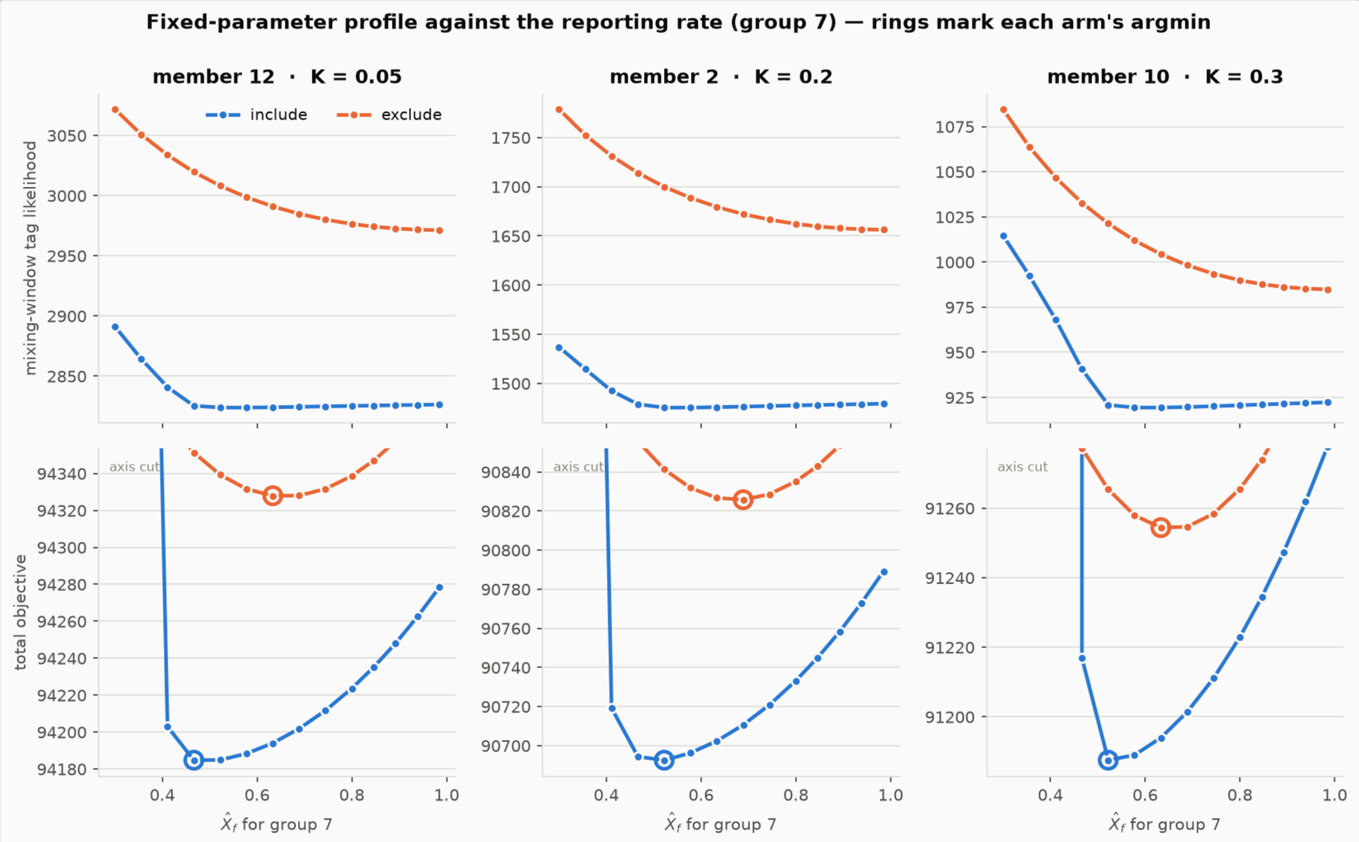
first differences of the exclude mixing block are negative in 100.0% of grid steps (worst profile 100.0%)

HELD `task1.include-mixing-block-invariant-to-X-above-0.8`

worst relative range of the include mixing block for X >= 0.8 = 0.0086; median 0.0009 -- this is the sub-range where the raised removal is small enough that the survival floor should not rescale the Newton-Raphson target

HELD `task1.penalty-is-identical-under-both-arms`

max |rr_penalty(exclude) - rr_penalty(include)| = 0 across the grid



Top: under **include** the mixing-window likelihood is flat in the reporting rate (0.09% median variation above X̂ = 0.8); under **exclude** it slopes toward X̂ = 1 in 100% of grid steps. Bottom: the best-fitting reporting rate sits **higher** under **exclude** in every member (+0.111 to +0.167 in X̂ for group 7).

The pre-registered stop condition was that the **include** curve be flat everywhere. Over the whole sweep it is not — up to 22.5%. The reason is the survival guard, and isolating it turns out to answer a question the SC22 review left open (Part 3).

DID NOT HOLD `task1.include-mixing-block-invariant-to-X`

worst relative range of the include mixing block over the full grid = 0.2254 (tolerance 0.02); median 0.0701

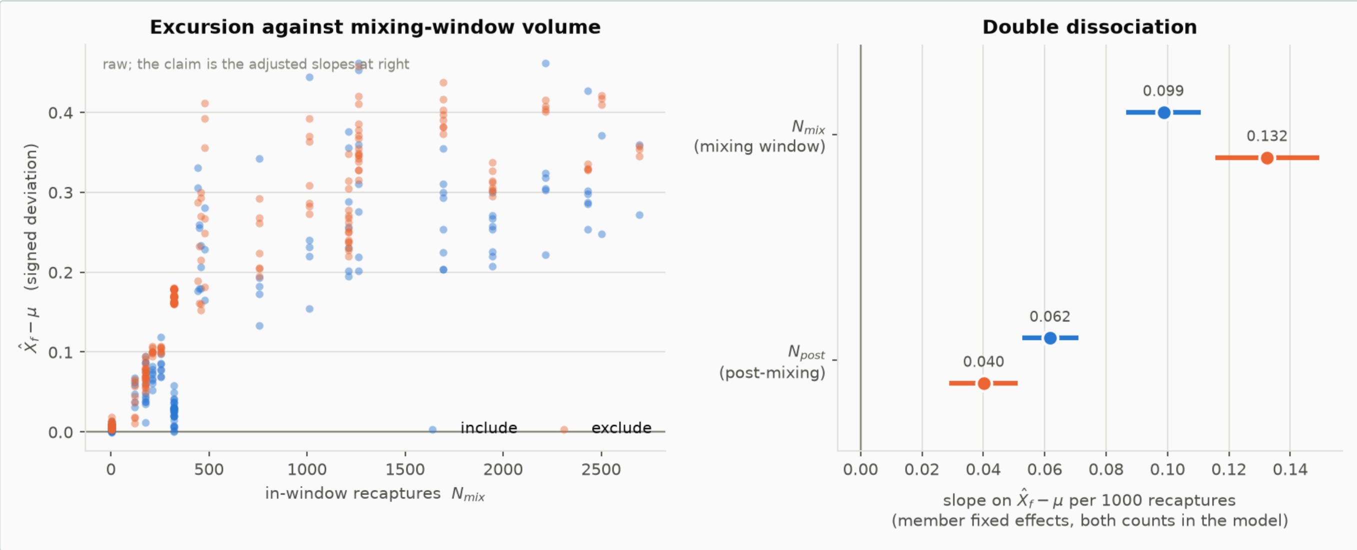
The argmin shift splits by prior weight, not by member: group 7 (weight 354.5) moves in all three members; group 17 (weight 739.2) is pinned by its heavier prior. That reproduces the group ordering seen in the ensemble (+0.142 vs +0.014) from an independent, fixed-parameter experiment.

HELD `task1.argmin-shift-larger-for-the-lower-weight-group`

median argmin shift: group 7 (w=354.5) +0.1667 vs group 17 (w=739.2) +0.0000 -- the same ordering as the observed ensemble arm differences (+0.142 vs +0.014)

What orders the pull across groups

It scales with the *count* of mixing-period recaptures resisted by prior weight — not the in-window *share*, which is why an earlier ψ̂-based test found nothing. Raw `N_mix` predicts nearly as well in both arms, which alone approaches the falsification threshold; the discriminating test controls for post-mixing returns, which carry the same identification benefit and none of the pseudo-data pull.



Under **exclude** mixing-window returns pull *harder* (+34%) while post-mixing returns pull *less* (−35%), both intervals excluding zero. That opposite movement is the signature of re-weighting toward pseudo-data.

HELD `task3.Nmix-and-Npost-arm-effects-have-opposite-signs`

N_mix slope 9.868e-05 -> 1.325e-04 (+34%) while N_post slope 6.177e-05 -> 4.014e-05 (-35%); both interaction CIs exclude zero

PART 3 · **RESOLVING AN OPEN QUESTION**

Which MFCL routine the assessment actually used

The SC22 review flagged an ambiguity it could not resolve from source:

> *"the source defines more than one version of the routine that performs this removal, and they differ both in whether they honour `tag_flags(i,2)` and in whether the cap is applied at all … we have not been able to confirm whether it is also the version carrying the cap."*

That is exactly right, and reading the current source does not settle it — it makes it worse. At `multifan-cl` HEAD the routine actually called is `do_newton_raphson_for_tags2`, which divides by the reporting rate **unconditionally** and applies **no cap**. The variant that honours the flag and carries the cap sits at `tag3.cpp:2769` and is never called; a third copy is commented out. The assessment binary is version **2.2.7.9** and predates that state of the source, so the source cannot answer the question — but the binary's own behaviour can.

Decompose the objective along a fixed-parameter profile. Everything except the tag terms is frozen, so

```
residual = objective − ( mixing-window tag block + post-mixing tag block + reporting-rate penalty )
```

is constant unless some *other* reporting-rate-dependent term exists. It is:

ARM	MAX UNEXPLAINED TERM	AT THE FITTED RATE	PEAKS AT
exclude	7.7e-07	0	— (flat everywhere)
include	190,673	1.04	$\hat{X}$ = 0.30, the largest raising

HELD `task10.no-unexplained-objective-term-under-exclude`

exclude: the objective is fully accounted for by the tag blocks and the reporting-rate penalty -- max unexplained range 7.65e-07 objective units across all profiles

HELD `task10.binary-carries-the-cap-and-honours-the-flag`

resolves the SC22 open question empirically: the 2.2.7.9 binary both honours tag_flags(i,2) (the arms differ) and applies the posfun cap (an unexplained penalty term appears under `include` at low X only)

The binary both honours the flag and applies the cap. The arms differ, so the flag is live; and an unexplained penalty appears under **include** only, growing as the raised removal grows, which no other term in the objective can produce. The residual above $\hat{X}$ = 0.8 is not quite zero below, which is why the cancellation is near-exact rather than exact in that sub-range.

DID NOT HOLD `task10.unexplained-term-vanishes-above-X-0.8`

include: unexplained term range above X = 0.8 is 2.7 -- the cap does not engage there, which is why the cancellation is exact in that sub-range

This also fixes the cost of the guard. The cap is what the manual's recommendation is designed to avoid — and at the reporting rates these models actually estimate it costs at most 1.04 objective units.

PART 4 · **THE DEEPER PROBLEM**

Both options bypass the likelihood

The SC22 review identifies a problem that both settings share, and it subsumes the mechanism above. Mixing-period recaptures fix the tagged population **deterministically**, from the observed returns, with no uncertainty attached and no test against what the model predicts.

Our source trace shows exactly how. Likelihood terms for these recaptures *are* formed — but their predictions are constructed from the observations rather than from the model. Under **include** the prediction reproduces the observation, so the term is inert. Under **exclude** the prediction is a reporting-rate fraction of the observation, and the only way to close that gap is to conclude reporting is complete.

So a substantial share of the tagging data steers the assessment without ever being fitted. Because the removal is deterministic, it carries *more* authority than fitted data would: a fitted observation can be outweighed by other information, a deterministic constraint cannot.

The influence does not stay inside the mixing window

It is worse than a contained problem, and this is a source fact, not an inference. `tag_fish_rep` — the estimated reporting rate $\hat{X}$ — is a single free parameter per (release group, fishery) cell, not one value per period. The likelihood-scoring loop in `src/tagfit.cpp` runs every period from release onward with no branch on it, `for (ip = initial_tag_period; ip <= ub; ip++)`, and applies the identical multiplier — `tag_rep_rate(it,ir,ip,fi) * tagcatch(it,ir,ip,fi)` — at each one, mixing and post-mixing alike; `tag_rep_rate` resolves to `tag_fish_rep(it, pp)`, a function of release group and fishery only, never of the period `ip`. So whatever pull the mixing-period pseudo-data puts on $\hat{X}$ under **exclude** is carried, completely unchanged, into the predictions the model compares against the genuinely-fitted post-mixing recaptures for that same cell — the parameter contaminated in the window it was meant to protect is the same parameter used to score real data outside it.

This is exactly what Part 2's `N_mix` / `N_post` double dissociation shows empirically: the same $\hat{X}$ responds to both mixing-period and post-mixing recapture counts at once, with the mixing-period component pulling harder under **exclude** and the post-mixing component pulling less — one parameter, contested between contaminated and genuine evidence, and the contaminated side wins more of that contest under **exclude**. Confining the damage to a few free parameters would already be a problem; parameters that are then used, unmodified, to score real data is worse.

This was a reasonable response to a real difficulty — tagged fish are not mixed, so predicting their recaptures from population-average availability is knowingly wrong, and declining to fit them is defensible. But unlike a data series that can simply be dropped, these fish have left the tagged population and still have to be removed from it. Desensitising the likelihood did not desensitise the model.

The one-sidedness argument

This is the strongest reason the axis does not belong in an uncertainty grid, and it needs no part of the channel-2 account. Assuming complete reporting **cannot mathematically remove more tags than correcting for reporting does**. The two options are one-sided. If estimated F rises with the ratio of recaptured to total tags — as it should — then assuming complete reporting imposes a *cap* on estimated F relative to the corrected option. Carrying both therefore widens the reported spread without bounding F or population scale from the direction that matters.

PART 5 · **CONSEQUENCES**

What it does to the advice

Stated on the 80 models meeting the assessment's convergence criterion (MGC ≤ 1e-4). That screen selects exactly the 80 members carrying a retained parameter file, so no conclusion changes coverage by switching between the two criteria.

HELD `task11.convergence-screen-selects-exactly-the-retained-par-set`

MGC <= 0.0001 selects 80 members and the retained-par set is 80; the two sets are identical, so no conclusion changes coverage by switching criterion

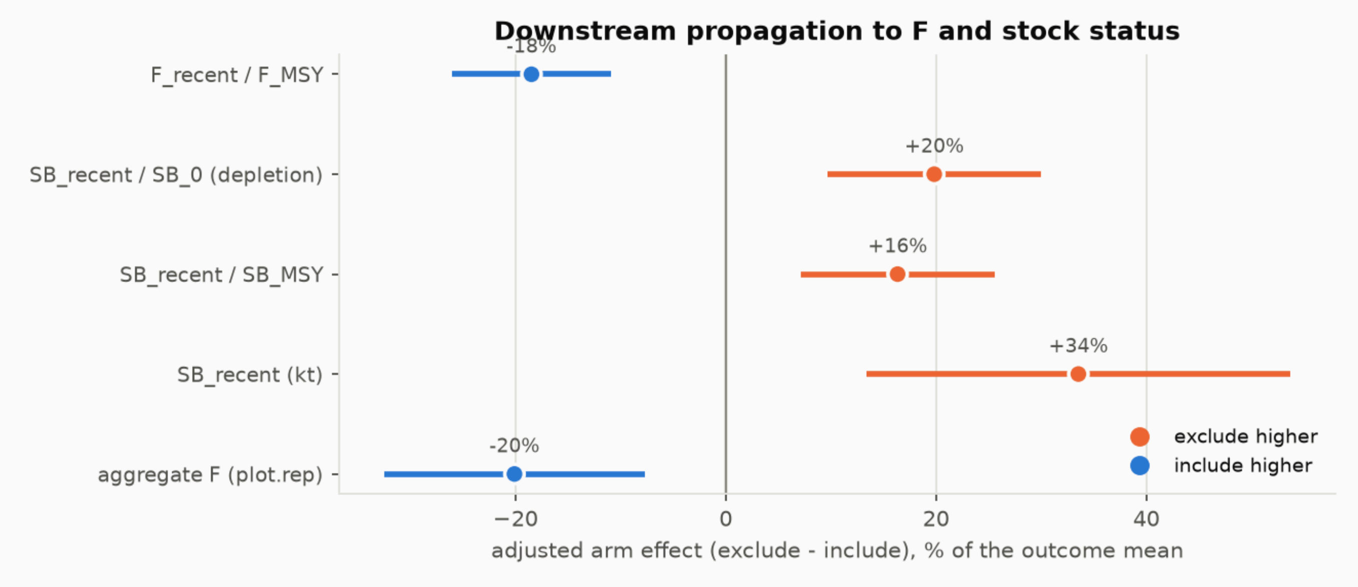
Adjusting for every other grid axis, **exclude** carries **-0.132** [-0.198, -0.066] on F/F_{MSY} — -15% of the ensemble mean — and **+0.0466** [0.0180, 0.0752] on depletion, +17%. Both are smaller than the same fits on all 88 models that ran (−0.166 and +0.053), because the convergence screen removes **include** models disproportionately.

HELD `task5.arm-effect-on-F-is-negative[f_recent_fmsy]`

F/F_MSY: -0.1664 [-0.2347, -0.09819], p=5.83e-06 (mean 0.9007)

HELD `task5.arm-effect-on-depletion-is-positive`

SB/SB_0: +0.05338 [0.02608, 0.08067], p=0.000204 (mean depletion 0.2692)



Every management quantity moves as the mechanism predicts and no interval crosses zero. This is the size of the effect, not evidence for which channel caused it — both channels push F the same way and compound.

An important negative result

We tried to decompose that effect cheaply, by measuring how much F moves per unit reporting rate with parameters frozen. It moves **not at all**: across 168 evaluations the objective took 56 distinct values per model while F/F_{MSY} took **1**, a range of exactly 0.

DID NOT HOLD `task9.fixed-parameter-profile-can-decompose-the-downstream-effect`

it cannot: dF/dX and the removal channel are both structurally zero at frozen parameters, so the observed -18% arm effect on F/F_MSY is an estimation effect and only re-estimation can decompose it

The reason is structural: the reporting rate enters only the tag likelihood, its own penalty and the tagged-cohort solver, and the tagged cohort is fitted *to* the population without feeding back into it. So the downstream effect is **optimiser behaviour** — the reporting rate reshapes the likelihood surface and the F-related parameters relocate during estimation. That rules out a class of explanation, and it removes the cheap instrument: only re-estimation can measure the pathway.

Fragility of the tagging module, and what it does to the comparison

The losses are heavily arm-specific, and they compound across two stages:

STAGE	INCLUDE	EXCLUDE
Drawn	50	50
Failed to build / run	9	3
Ran but failed MGC ≤ 1e-4	7 (17%)	1 (2%)
Surviving	**34**	**46**
90th-percentile max gradient	7.7e-04	9.9e-05

HELD `task11.tagging-module-is-arm-specifically-fragile`

cumulative loss from the designed 50 draws: include 16/50 (32%) vs exclude 4/50 (8%) -- a factor of 4.0

HELD `task11.convergence-failure-alone-is-arm-specific`

of models that ran: include 7/41 (17.1%) failed MGC vs exclude 1/47 (2.1%)

The convergence screen therefore *degrades* the randomisation the design bought. The **include** share falls to 42%, and the standardised arm differences on the other axes widen — M0 to -0.211, effort creep to +0.184, K to +0.111.

HELD `task11.convergence-screen-worsens-arm-balance`

include share falls from 46.6% to 42.5%; |SMD| on K goes 0.044 -> 0.111

Does the imbalance understate the difference?

Yes for the raw comparison, and no for the adjusted one — and the distinction matters.

The **raw** arm difference in F/F_{MSY} falls from -0.1415 across all models that ran to **-0.0858** on the converged set — a 39% shrinkage. Imputing the lost members at their own design points and completing both arms to the designed 50/50 restores it to -0.1264. The lost **include** models sit where F is high, so removing them pulled the **include** mean down and closed the gap.

HELD `task11.raw-gap-is-materially-truncated-by-the-convergence-screen`

raw F/F_MSY gap falls from -0.1415 (all that ran) to -0.0858 (converged only), a 39% shrinkage; completing the arms restores it to -0.1264

The **adjusted** estimate barely moves. Three routes to a balanced representation — reweighting to the designed arm × K marginal, completing both arms by imputation, and a genuinely balanced **subset of models that actually ran** (below) — give F/F_{MSY} arm effects spanning -0.142 to -0.135, against -0.132 unbalanced.

HELD `task11.adjusted-gap-is-robust-where-the-raw-gap-is-not`

adjusted estimate moves +2.4% on completion while the raw difference moves +47% -- the adjustment already conditions on the axes that drive the differential loss, so the adjusted effect is the one to quote

That is the expected pattern rather than a coincidence: the adjustment already conditions on the axes along which the losses are concentrated, so the composition shift that moves the raw difference is precisely what the adjusted model absorbs. **The practical consequences are that raw arm-versus-arm comparisons on the retained ensemble understate the difference by roughly 40%, and that the adjusted estimate is the one to quote.** A more balanced ensemble would change the headline little; it would change any unadjusted presentation a great deal.

A balanced subset of real runs, not interpolation

The routes above lean on a fitted model to stand in for the composition the design intended. It is worth also asking the question with no model at all: is there a subset of the 80 converged runs, *as fitted*, that is balanced on its own? Optimal 1:1 matching — pairing each **include** model to the **exclude** model closest to it on the five continuous axes (K, τ, M0, effort creep, h), by minimum total distance, with a caliper to drop poor matches — finds **27 pairs** (27×2 of the 80 converged models) with every axis balanced to |SMD| < 0.15.

HELD `task11.matched-pairs-achieve-good-covariate-balance`

worst |SMD| across (K, tau, M0, creep, h) after optimal 1:1 matching is 0.150 on 27 pairs (caliper 1.627 std units, 7 candidate pairs dropped as poorly matched)

On those 27 pairs, the **raw, unadjusted, unmodelled** paired difference is F/F_{MSY} -0.162 [-0.234, -0.090] and depletion 0.0560 [0.0298, 0.0821] — matching or exceeding the adjusted whole-sample estimate, using no regression at all. This is the cleanest evidence that the imbalance was suppressing the raw comparison rather than the adjustment manufacturing an effect that is not really there.

HELD `task11.raw-paired-difference-corroborates-the-adjusted-estimate`

on 27 real, covariate-matched pairs the RAW (unadjusted, no model) paired difference is -0.1621 [-0.2342, -0.0899], at least as large as the adjusted whole-sample estimate (-0.1315) despite using no regression at all

The reweighting and imputation routes rest on an assumption worth stating: that within a design cell the retained members stand in for the lost ones. Differential convergence is exactly what makes that doubtful, so those two are illustrative bounds, not corrections — the imputation in particular extrapolates to the design points where estimation failed, where a smooth model of the outcome is least trustworthy. The matched-pairs result does not carry that assumption: it uses only models that were actually fitted.

What these numbers do not include

Every risk and arm-effect figure in this document — the adjusted estimates, the matched pairs, the $P(F>F_{MSY})$ and P(SB<LRP) figures above — is built from each model's single best-fitting parameter vector, one point per grid cell. None of it propagates **estimation uncertainty**, the spread of parameter values consistent with the data within a single fit. What is reported here is the **structural** component only: how the point estimate moves across the grid. That is a real, distinct restriction, not a formality — the repository's own combined structural-and-estimation audit (`results/tables/estimation-uncertainty-audit.csv` , pooling 100 joint-Hessian draws per model with equal weight

across all 80 retained models) puts the 10th–90th percentile of SB_{recent}/SB_{MSY} at roughly 0.82 to 2.57 around a median of 1.53 — a band wide enough that estimation uncertainty is not a minor addition on top of the structural spread reported here. That table is pooled across **both arms together**, so it confirms the scale of what is missing without being able to say how the two arms' risk figures would move relative to each other.

Answering that would need the per-model draws split and recombined by arm, and the repository only partially supports doing so. Per-model joint-Hessian draws exist for **62 of the 80 converged models** (25 of 34 **include**, 37 of 46 **exclude** — the 18 gaps are exactly the models without a positive-definite Hessian, split evenly across arms, so not themselves an added source of imbalance). Even with full coverage, combining a hundred joint draws per model into a single arm-conditional risk estimate is a real undertaking, not a quick addition, and the existing pipeline for it (`report/augment-uncertainty-projection.R`) requires an R toolchain with several packages that this analysis environment does not have. It was not attempted here. **The structural-only figures above should be labelled as such wherever they are used, and should not be read as the full assessment uncertainty on the arm comparison.**

PART 6 · **THE CHOICE**

Both options are flawed. They are not equally flawed.

EXCLUDE — no reporting-rate correction

- **Assumes complete reporting.** The model's own fitted rates contradict this for most of the tagging data.
- **Pseudo-data on 96 of 98 release groups**, in every model. The in-window terms are minimised at $\hat{X} = 1$ regardless of the data.
- **Known directional bias.** It can never remove more tags than the corrected option, so wherever reporting is incomplete it leaves too many tags in the population — lower F, larger population scale.
- **Estimated rates pushed up**, systematically closer to the 0.99 bound.
- **Diagnostics disagree with the fit** by ~21.6% in the mixing window.
- **Numerically well-behaved** — 6% of draws failed, no cap engagement.
- **What the MFCL manual recommends.**

INCLUDE — correct returns by $\hat{X}$

- **Capable of being unbiased**, under three conditions: the reporting rate is about right, release groups are large enough, and the raised removal leaves enough tags.
- **No pseudo-data in the general case** — $\hat{X}$ cancels out of the in-window terms.
- **The third condition is the fragile one.** The cap engages on 2 of 98 release groups, and where it does the same $\hat{X}\cdot R$ form partially returns. Nothing in standard output reports it.
- **Small release groups are a real risk** — SC22 simulations show a heavy upper tail in F rather than an obvious loss of precision.
- **The cap penalty is discarded outright** for release groups 21, 72 and 112 (hardcoded), and 21 is one that engages.
- **Numerically fragile** — 32% of the 50 draws lost against 8% for **exclude**, counting build failures and convergence failures together. Informative non-response, not random loss.

Both settings estimate reporting rates above the external evidence

The two options are sometimes described as bracketing the reporting-rate adjustment: not correcting is equivalent to assuming RR = 1, which applies the smallest possible raising factor of 1.00, while correcting applies $1/\hat{X} > 1$. That framing holds only if $\hat{X}$ is exogenous. It is not — $\hat{X}$ is estimated inside the same likelihood, and in this assessment it lands well above the tag-seeding evidence in **both** arms.

PRIORED GROUP	SEEDING-TRIAL RR	FITTED, INCLUDE	FITTED, EXCLUDE	RAISING APPLIED, WIDEST OPTION	RAISING IMPLIED BY SEEDING
7 · RTTP PS.2	0.496	0.524	0.666	1.91	**2.02**
10 · RTTP PS.WEST.3	0.512	0.579	0.588	1.73	**1.95**
14 · PTTP PS.2	0.496	0.502	0.502	1.99	**2.02**
17 · PTTP PS.WEST.3	0.512	0.768	0.783	1.30	**1.95**
18 · PTTP PS.EAST.3	0.528	0.815	0.900	1.23	**1.89**

In **5 of 5** priored groups the raising implied by the seeding trials exceeds what either option applies. For groups 17 and 18 — the two that dominate the penalty — the widest option raises by 1.30 and 1.23 against seeding-implied factors of 1.95 and 1.89.

The pattern is diagnostic rather than incidental. Group 14 is the one case where the applied raising nearly reaches the seeding value (1.99 against 2.02) — and group 14 has **six** in-window recaptures, too few to move its rate off the prior. Where the tag data is abundant enough to move $\hat{X}$, $\hat{X}$ rises and the gap to the external evidence opens. So the span between the two options is a span in the *fitted* adjustment, and the fitted adjustment is itself displaced upward relative to the independent seeding estimates.

Removing the feedback into the tagged population does not remove the influence on the fit

A natural defence of the uncorrected setting is that it avoids circularity: with no correction, none of the parameters estimated from the tag likelihood influence the tagged population abundance at the end of the mixing period, so that abundance — which drives all post-mixing predictions — is not contaminated by the reporting rate. That is true as far as it goes, and it is the design intent.

But it tracks influence in one direction only. The mixing-period *likelihood terms still exist* under both settings, and under the uncorrected setting they contain $\hat{X}$ while their predictions are built from the observed returns. So the mixing-period data continues to influence $\hat{X}$ even though $\hat{X}$ no longer influences the tagged population. And $\hat{X}$ is not confined to the mixing period: it scales every predicted recapture *outside* the window as well. The path runs mixing-period returns → $\hat{X}$ → post-mixing predictions → the tag likelihood, and it is not closed by breaking the feedback into the cohort.

The circularity is therefore cut in one direction and left open in the other, and the direction left open is the one that carries the bias — because those in-window terms are minimised at $\hat{X} = 1$ regardless of what the data say.

The recommendation

If advice must come from models fitted to tagging data, it should come from include models only — and the reporting-rate axis should be removed from the uncertainty grid rather than rebalanced.

Three independent reasons, none of which requires accepting the whole channel-2 account:

- **One-sidedness.** The options do not bracket. Carrying both cannot bound F or population scale from the direction that matters, so the spread it produces is not uncertainty.
- **Asymmetric capability.** **Include** can be unbiased under stated conditions. **Exclude** is biased low unless mixing-period reporting is genuinely complete, and it is not.
- **Asymmetric contamination.** Established by source transcription and a fixed-parameter experiment: pervasive under **exclude** (96 of 98 groups), localised under **include** (2 of 98).

The manual's recommendation is a bad trade

The manual prefers **exclude** because the correction can be numerically unstable. That concern is real but small, and we have now measured both sides of the trade:

	RELEASE GROUPS AFFECTED	COST AT FITTED RATES
The cap (**include**'s problem)	2 of 98	≤ 1.04 objective units
Pseudo-data (**exclude**'s problem)	96 of 98	shifts F/F_{MSY} by -15%

About **48 times** as many release groups are affected by the problem the manual's advice creates as by the problem it avoids. The guard fires only where the raised removal is extreme — at reporting rates far below any the models estimate — and where it does fire it costs a fraction of an objective unit. **The manual is optimising for numerical convenience at the cost of a known directional bias in the answer.**

Two honest qualifications

Include's higher failure rate is *informative* non-response: the models that failed are those where the correction was least feasible, so the surviving set is conditioned on the correction being benign. And neither arm's rates should be treated as truth — the external anchor is the tag-seeding priors, not either fit.

An **include**-only ensemble already exists and needs no new runs: the 34 retained members cover every level of every other axis. Acting on this needs a decision, not compute.

PART 7 · SUPPORTING FINDINGS

The rest of the evidence

The diagnostic defect

The tag-fit panels circulated with assessments cannot detect misfit inside the mixing window. The report writers apply a different multiplier from the likelihood: in-window they use 1.0 under **exclude** while the objective uses $\hat{X}$. Since the solver drives the in-window predicted catch to the observed returns, the plotted fit is an **identity in both arms** — verified against the circulated `plot.rep`, where time-at-liberty bin 1 gives pred/obs of 1.0000–1.0207 across members of both arms, against 0.80–0.83 in the first fully post-mixing bin.

HELD `task2.report-likelihood-discrepancy-still-holds`

exclude arm: the objective's in-window predicted returns sit 21.6% below the values written to plot.rep (median 802 of 3605 fish)

Under **exclude** the objective's in-window predictions sit **21.6%** below the plotted ones — 802 of 3605 fish per model. This is an MFCL defect worth reporting upstream regardless of which setting is adopted.

The reporting-rate bound

Bound contact (at 0.99) occurs in two **include** members and none under **exclude**, which looks backwards. Re-testing on the estimation scale — an arcsine-of-log transform, so equal proportion gaps are not equal distances — does not reverse the ordering, but the distribution resolves it: **exclude** sits systematically closer to the bound in median (transformed gap 308.6 vs 359.6), and the two **include** contacts exceed the entire **exclude** distribution. Contact counts were the wrong statistic.

HELD `task4.exclude-closer-to-bound-across-all-priored-groups`

median transformed gap over all priored groups: include 359.6 vs exclude 308.6

DID NOT HOLD `task4.bound-contact-test-in-transformed-space`

proportion scale: include 5.9% vs exclude 0.0%; estimation scale: include 5.9% vs exclude 0.0% -- ordering does not reverse

The survival floor across the ensemble

A `bet.tag` -based screen (deliberately conservative) flags floor engagement in 100% of **include** members and none under **exclude** — consistent with the objective decomposition in Part 3. Engagement concentrates in the pooled release group (98) and group 21, whose penalty MFCL discards.

HELD `task8.floor-engagement-higher-under-include`

share of members with >=1 engaging release group: include 100.0% vs exclude 0.0%

HELD `task8.lowest-penalty-include-members-are-more-floor-engaging`

lowest-penalty tercile 2.00 vs highest 1.36 engaging release groups; Spearman(penalty, n_engaging) = -0.588

$\hat{\psi}$ under three anchors

$\hat{\psi}$ inherits the rates it is raised by, so the published diagnostic — computed with **exclude**-arm rates — is itself subject to channel 2. Recomputing under the externally anchored tag-seeding priors raises release-weighted $\hat{\psi}$ from 0.409 to 0.414 (+1.1%), consistently in 80 of 80 members. **But the correction is second-order:** the mixing configuration moves $\hat{\psi}$ by 0.563 across the K axis, about 120 times more than the raising scheme does.

HELD `task6.psi-higher-under-external-anchor-than-as-published`

release-weighted psi-hat: as published (exclude rates) 0.4091 -> external anchor (seeding priors) 0.4137 (+1.1%)

HELD `task6.mixing-configuration-dominates-the-raising-scheme`

psi-hat moves 0.5635 across the K axis (from 0.7860 at K=0.05 to 0.2225 at K=0.35) versus 0.0047 between raising schemes -- a factor of 120

The published baseline does not reconcile: the closest configuration (K = 0.30) gives $\hat{\psi}$ = 0.226 against the published 0.231, but 35 groups at $\hat{\psi} \geq 0.3$ against 27. These tables are therefore **not** drop-in replacements; the scheme contrast is internally consistent, the level is unreconciled, and that gap should close before anything circulates.

DID NOT HOLD `task6.published-baseline-reconciles`

closest configuration is K=0.3: psi-hat 0.2259 vs published 0.231 (close), but 35 groups at psi>=0.3 vs published 27 (does not reconcile)

PART 8 · **STATUS AND NEXT STEPS**

What is established, and what is not

Established by transcription or controlled experiment

- The penalty's exact functional form, reproduced to 4.9×10^{-10}.
- That **exclude**'s mixing-period terms are minimised at $\hat{X}$ = 1, and **include**'s are inert in $\hat{X}$ except where the cap engages.
- That flipping the flag alone, at frozen parameters, moves the best-fitting reporting rate up.
- That the assessment binary both honours `tag_flags(i,2)` and applies the cap — resolving an ambiguity the source at HEAD cannot.
- That the reporting rate cannot move F at frozen parameters; the downstream effect is optimiser behaviour.
- That the in-window tag-fit diagnostic is an identity in both arms.

Established observationally, on a selected non-crossed ensemble

- Adjusted axis effects on the 80 converged models: +34.3 [27.0, 41.7] on the penalty, -0.132 on F/F_{MSY}, +0.0466 on depletion.
- The `N_mix` / `N_post` double dissociation.
- That **exclude** sits systematically closer to the reporting-rate bound.
- That the **include** setting is arm-specifically fragile — 32% of drawn models lost to build or convergence failure against 8% for **exclude** — and that on 27 covariate-matched pairs of real, converged runs the raw F/F_{MSY} difference (-0.162) matches the adjusted whole-sample estimate without any model behind it.

Not established

- **That the downstream effect is caused by the reporting rate.** Both channels push F the same way and compound; conditioning on $\hat{X}$ over-mediates because $\hat{X}$ and F are jointly estimated. Only paired refits can settle it.
- **That either arm's rates are correct.**

- **A reconciled $\hat{\psi}$ correction.**
- **The magnitude of the small-release-group risk** in this assessment specifically — SC22 simulation indicates a heavy upper tail in F, but we have not quantified it here.
- **How the risk figures change once estimation uncertainty is included.** Every P(F>F_{MSY}) and P(SB<LRP) figure here reflects structural uncertainty only — the spread of single best-fitting parameter vectors across the grid — with none of the within-model parameter uncertainty folded in. The ensemble's own audit shows that uncertainty is large in scale; it is pooled across both arms in the repository and was not decomposed by arm here (Part 5).

Recommended sequence

1. **Report the diagnostic defect upstream.** No compute, and independent of the include/exclude decision.
2. **Take the axis out of the grid.** Present **include**-only, or present the arms separately with the one-sidedness stated. Presenting a 20%-versus-29% split in limit-exceedance probability as uncertainty about the stock is not defensible.
3. **Reconcile the $\hat{\psi}$ baseline** before any corrected figure circulates.
4. **Then paired refits** — the only instrument that can measure the downstream pathway, now that the fixed-parameter route is closed. Time one full-path run before committing to a count, and consider targeting the 9 failed **include** members, since recovering them removes the selection concern on an **include**-only ensemble.

The durable fixes

Both come from the SC22 review and neither is a grid adjustment.

1. **Modify MFCL** so the mixing-period likelihood terms that already exist score predictions generated from the projected tagged cohort, with mixing-period availability and the reporting rate estimated rather than assumed. This follows the original design reasoning instead of discarding it — the obstacle was that recaptures cannot be predicted from an unmixed cohort, so estimate the departure from mixing. **This adds parameters to a model that already has an identifiability problem, not a clean one.** The same review's own diagnostics — jittering, simulation self-test, parameter correlations, bounds checks — found that several parameters of the 2026 assessment's spatial structure are "poorly determined and weakly identifiable" before any availability parameters are added. Layering one more per-release-group parameter onto that model, with tag release groups that are mostly small and noisy, could compound the existing estimability problem rather than trade it for a cleaner one. SC22 says as much of its own proposal: "it remains to be seen if the additional parameters would be identifiable." There is also a resolution requirement independent of that: an availability multiplier too coarse to span the true departure from mixing inflates mixing-period mortality and biases F upward instead of removing the bias it targets. **This is a research direction pending its own simulation testing — not a fix ready to adopt — and that testing should establish identifiability before implementation is considered, not after.**
2. **Include spatially implicit, fleets-as-areas models** in the grid, which sidesteps the tagging treatment entirely and integrates over spatial-structure uncertainty as well.

Given both of those carry open questions — one on identifiability, the other a simplification of the population dynamics — a third, more pragmatic option follows directly from everything above: **stop fitting tag recoveries inside MFCL's joint likelihood and treat them as an external analysis instead** — an independent tag-based estimate of F or natural mortality used as a cross-check or to inform a prior, rather than a term optimised jointly with the rest of the model. This sidesteps the channel-1/channel-2 pathology in Part 2 and the added-complexity risk above without waiting on either fix, at the cost of losing the tagging data's direct influence on the fit. It is not a new SC22 recommendation — it generalises an aside in their second path, that a model not fitting tagging data directly can still use it "to inform prior distributions for mortality parameters" — to the current spatially-explicit structure rather than tying it to a change in spatial resolution.

Analysis code: `analysis/rr-penalty/`, `analysis/channel2/` and `analysis/summary/`, reproduced by the `run-all.sh` in each. Tidy CSVs, figures and the full check ledgers in `out/` and `out/channel2/`. Source excerpts pinned to `multifan-cl` commit `624dee4f`; assessment binary version 2.2.7.9 — note the two differ, which Part 3 addresses. Qualitative framing, the one-sidedness argument, the three conditions for the corrected option and the two paths forward are drawn from N-DucharmeBarth-NOAA/sc22-tag-analysis (commit `907274f`). Objective components and the design table cover all 88 retained models; 80 have a retained `final.par`. No estimation run was launched for either investigation — every MFCL invocation set `parest_flags(1) = 0`.